

Original Article

Using machine learning to predict patients with polycystic ovary disease in Chinese women



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ARTICLE INFO

Article history:

Accepted 10 September 2024

Keywords:

Machine learning

Logistic regression

Polycystic ovary syndrome

ABSTRACT

Objective: With an estimated global frequency ranging from 5 % to 21 %, polycystic ovary syndrome (PCOS) is one of the most prevalent hormonal disorders. There are many factors found to be related to PCOS. However, most of these researches used traditional methods such as multiple logistic regression (LR). Nowadays, machine learning (Mach-L) emerges as a new method and can be used in medical researches. In the present study, there were two goals: 1. Compare the accuracy of five alternative Mach-L techniques with that of conventional LR. 2. Use Mach-L to forecast PCOS and prioritize the risk factors. **Materials and methods:** Totally, 170 PCOS patients and 950 control participants were included. We collected information on demographics, biochemistry, and lifestyle. PCOS was identified using Rotterdam criteria. Random Forest (RF), stochastic gradient boosting (SGB), multivariate adaptive regression splines (MARS), extreme gradient boosting (XGBoost), and gradient boosting with categorical features support (CatBoost) are five Mach-L algorithms that were used. Models with lower estimation errors were better. **Results:** By using *t*-test, we found subjects with PCOS were younger, glutamic oxaloacetic transaminase (GOT), glutamic pyruvic transaminase (GPT), γ -Glutamyl transferase (γ -GT), Triglyceride (TG), and educational levels were higher. All the five Mach-L methods had lower estimation errors compared to LR. The average of the AUC derived from Mach-L was mean AUC of 0.6669, higher than the that of LR (0.5908). Finally, age, TG, GPT, white blood cell count (WBC), uric acid (UA), and platelet (Plt) were the six most important risk factors selected by Mach-L.

Conclusion: Mach-L methods overtook conventional LR and age was the most significant factor, followed by TG, GPT, WBC, UA, and Plt in a cohort of Chinese women.

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Introduction

Polycystic Ovary Syndrome (PCOS) is one of the most common endocrine disorders affecting women of childbearing age, with an estimated global prevalence ranging from 5 % to 21 % [1,2]. PCOS presents with a variety of symptoms and conditions, including

irregular menstruation, anovulation, infertility, acne, hirsutism, obesity, hyperlipidemia, and cardiovascular diseases [1,3].

Other than those, PCOS is also associated with various cardiovascular and metabolic abnormalities, including obesity, dyslipidemia, hypertension, prediabetes, or type 2 diabetes.

It should be noted that most of these previous studies have utilized traditional statistical methods such as multiple logistic regression (Logi-R) or multiple linear regression. To our knowledge, in the last 5 years, only 9 studies have employed machine learning techniques to analyze PCOS. Some of these studies focused on comparing different machine learning methods (Mach-L) or examining PCOS-related gene testing [4–6].

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Recently, the emerge of the artificial Intelligence based on Mach-L changes the landscape of medical researches. Mach-L can learn from past data or experiences without the need for explicit programming. It has two unique characteristics compared to LR: 1. It does not need hypothesis such as normal distribution; 2. It can capture non-linear relationships and complex interactions among the data, Mach-L has the potential to surpass Logi-R in disease prediction [7].

In the present study, we recruited 1120 women, our aims were: 1. Compare the accuracy of five different Mach-L methods with LR. 2. Using Mach-L to identify the most important risk factors for PCOS.

Materials and methods

Participants and study design

Some of the following contexts were published in our previous study [8]. The data for this study were derived from the Taiwan MJ cohort, which is a prospective health examination cohort conducted by the Taiwan MJ Health Check Center [9,10]. Each participant completed a self-administered questionnaire, which collected information on personal and family medical history, current health status, lifestyle, physical exercise, sleep habits, and dietary habits, among other details [10,11]. The MJ Health Database only includes participants who provided informed consent. The study protocol was approved by the Institutional Review Board (IRB) of Tri-Service General Hospital, National Defense Medical Center (IRB No: A202205013). A total of 2300 females were initially enrolled. After excluding participants for various reasons, 1120 participants were included in the analysis (Fig. 1). Among them, 170 participants were diagnosed with PCOS.

Measurements of anthropometry and biochemistry

On the day of the study, a senior nurse recorded the participants' medical history, including information about current medications. Subsequently, a physical examination was conducted. Body fat

percentage (BF) was measured using methods such as bioelectrical impedance analysis (BIA). WBC, hemoglobin levels, and platelet count (Plt) were measured using standard laboratory techniques, typically performed on automated hematology analyzers. Creatinine and uric acid (UA) levels were measured using blood tests that assess the concentration of these substances in the blood. CRP levels were measured using high-sensitivity assays that detect low levels of CRP in the blood.

The measurement of fasting plasma glucose (FPG), and lipid profile could be referred to our previous study [8].

Lifestyle information

Information on smoking, drinking, exercise time, sleeping time, and education level was obtained through questionnaire surveys. The drinking area was defined as the multiple of total drinking duration, frequency of drinking and the percentage of alcohol. Similarly, the smoking area was the multiple of the duration, frequency of smoking and number of cigarettes. The sport area was the multiple of duration, frequency, and type of the exercise. The sleep time was an ordinal variable as shown in Table 1.

Traditional statistics

The data is presented as a mean ± standard deviation. The student's t-test was used to assess the differences in continuous data between Without PCOS and With PCOS. All statistical tests were two-tailed, and $p < 0.05$ was considered statistically significant. Statistical analysis was conducted using SPSS 10.0 for Windows (SPSS, Chicago, Illinois, USA).

The definition of PCOS

According to the consensus reached by Dutch experts in 2003, the diagnosis of PCOS includes at least two out of the following three criteria: 1. Clinical or biochemical signs of hyperandrogenism,

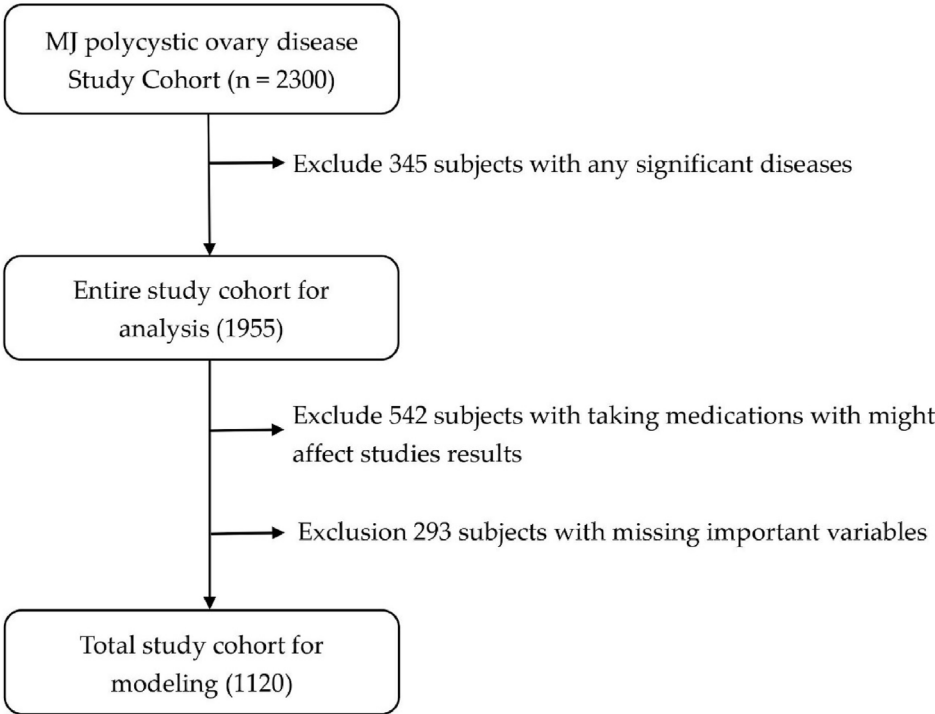


Fig. 1. Patient selection scheme.

Table 1
Demographic, biochemistry, and life style data of study cohort.

	Without PCOS	With PCOS
N number	950	170
Age (year)	32.770 ± 7.862	31.050 ± 5.518**
Body fat (%)	29.891 ± 8.115	29.670 ± 7.999
White blood cell (10 ³ /uL)	6.097 ± 1.703	5.960 ± 1.418
Hemoglobin (g/dL)	13.263 ± 0.989	13.392 ± 0.906
Platelet (10 ³ /uL)	260.270 ± 57.824	262.010 ± 60.501
Fasting plasma glucose (mg/dL)	94.520 ± 12.813	94.280 ± 8.883
Serum glutamic oxaloacetic transaminase (U/L)	20.260 ± 10.546	22.270 ± 10.972*
Serum glutamic pyruvic transaminase (U/L)	19.860 ± 17.646	25.790 ± 28.240***
γ-Glutamyl transferase (U/L)	17.660 ± 19.896	21.510 ± 19.692*
Creatinine (mg/dL)	0.809 ± 0.106	0.818 ± 0.118
Uric acid (mg/dL)	4.905 ± 1.134	5.077 ± 1.283
Triglyceride (mg/dL)	83.700 ± 47.995	93.420 ± 56.821*
High density lipoprotein cholesterol (mg/dL)	62.320 ± 14.925	60.940 ± 15.762
Low density lipoprotein cholesterol (mg/dL)	107.100 ± 30.585	107.860 ± 28.278
Thyroid stimulating hormone (uIU/mL)	1.690 ± 1.088	1.697 ± 1.053
C-reactive protein (mg/dL)	0.209 ± 0.329	0.191 ± 0.197
Education		
Illiterate	7 (0.74 %)	0 (0.00 %)
Primary school	20 (2.11 %)	1 (0.59 %)
Junior high school	19 (2.00 %)	1 (0.59 %)
Senior high school	202 (21.29 %)	25 (14.71 %)
College	215 (22.66 %)	49 (28.82 %)
University	381 (40.15 %)	75 (44.12 %)
Higher than master degree	105 (11.06 %)	19 (11.18 %)
Drinking	1.018 ± 5.481	1.018 ± 4.838
Smoking	0.886 ± 5.711	0.965 ± 6.421
Sport hour/week	4.400 ± 5.830	4.790 ± 7.470
Sleep time/day	3.070 ± 0.751	3.030 ± 0.629

PCOS: polycystic ovary syndrome, p: * < 0.05, ** < 0.01, *** < 0.001.

2. Persistent anovulation, and 3. The observation of polycystic ovarian morphology, while excluding any potential secondary causes [12].

Description of the study data set

The dependent variables collected from our participants are listed in Table 1. All demographic, biochemical, and lifestyle data are independent variables (continuous, ordinal, and categorical), while the presence of PCOS is the dependent variable (categorical).

Proposed scheme

This study proposed a scheme based on one traditional method (Logistic Regression, LR) and five Mach-L methods:

1. Random Forest (RF) method, which utilizes an ensemble technique to aggregate a large number of unpruned decision trees [13].
2. Stochastic Gradient Boosting (SGB) is an additive regression tree method that combines both bagging and boosting techniques [14].
3. Multivariate Adaptive Regression Splines (MARS) is a non-parametric regression technique based on a combination of piecewise polynomials [15].
4. Extreme Gradient Boosting (XGBoost) is an ensemble learning method that combines boosting and decision trees principles [16].
5. CatBoost (Categorical Boosting) is a gradient boosting algorithm that utilizes an ordered method for decision tree fitting and supports categorical features [17].

Fig. 2 illustrates the flowchart of the proposed approach combining five Mach-L methods. In the proposed scheme, we first collect patient data to prepare the dataset for model construction. The dataset is randomly divided into 80 % training data for model building and 20 % testing data for out-of-sample testing. During the training process, each Mach-L method has its own hyperparameters that need to be tuned to build well-performing models. We use the 10-fold cross-validation technique for hyperparameter tuning. For this purpose, the training data is further randomly split into training data for constructing models using different sets of hyperparameters and validation data for model validation. All possible combinations of hyperparameters are explored through grid search. The model with the lowest root mean square error on the validation data is considered the best model for each Mach-L method. LR, RF, SGB, MARS, XGBoost, and CatBoost's best models are generated, providing information on the corresponding variable importance rankings.

During the testing process, the testing dataset was used to evaluate the predictive performance of the best-performing models, including RF, SGB, MARS, XGBoost, and CatBoost. The evaluation of model performance was based on various metrics, such as accuracy, sensitivity, specificity, and area under the curve (AUC), as shown in Table 2. To provide a more robust comparison, LR was used as the benchmark model for comparing model performance. Hence, Mach-L models with higher AUC values than the LR model are considered convincing models.

Since all the Mach-L methods used can generate importance rankings for each predictor variable, we defined the priority in each model as follows: Rank 1 represents the most critical risk factor, and Rank 21 represents the least important risk factor. Different Mach-L methods may produce different variable importance rankings due to their unique modeling features. To enhance the stability and comprehensiveness of risk factor importance

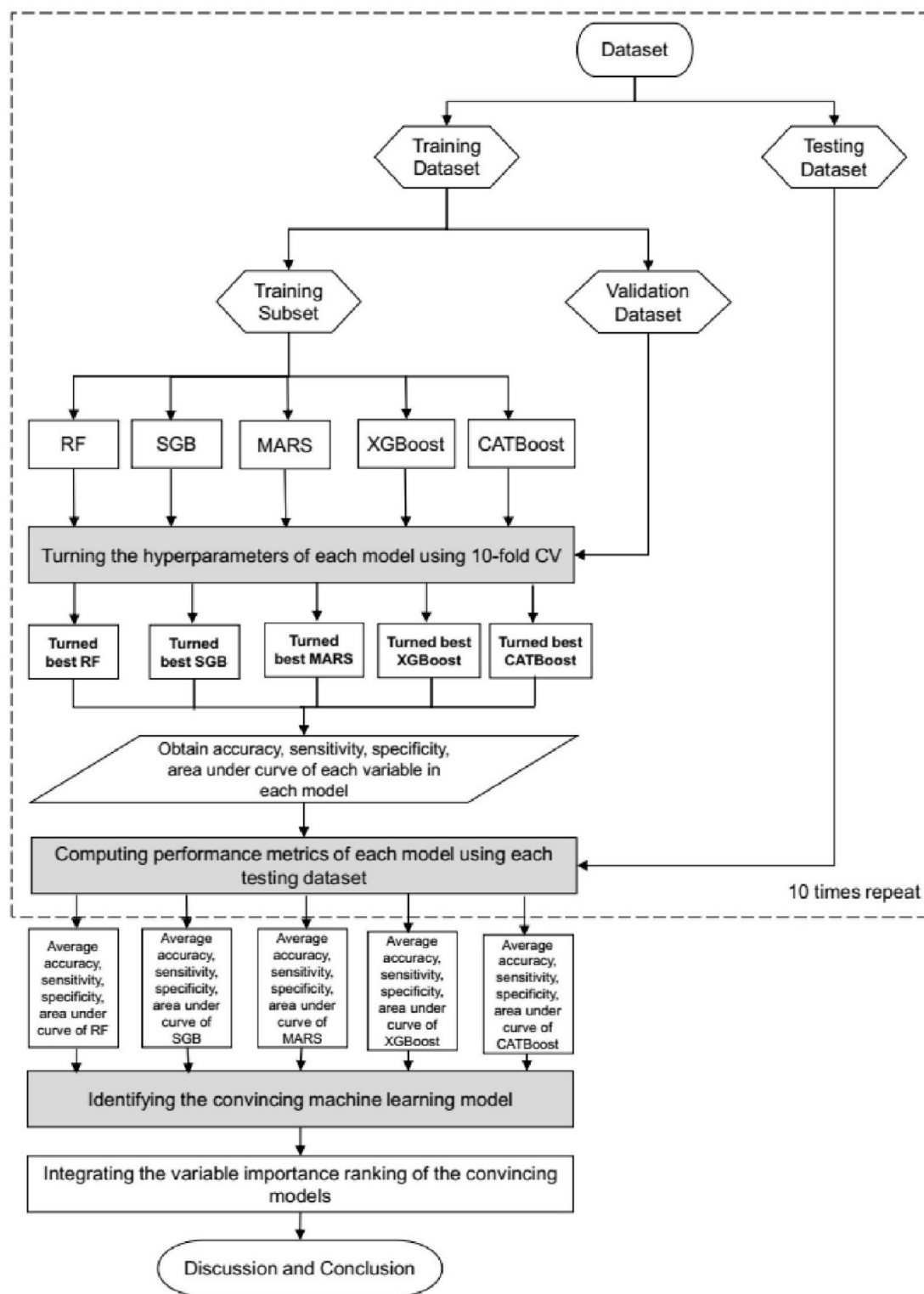


Fig. 2. Proposed scheme for five machine learning methods.

reordering, we integrated the variable importance rankings from the convincing Mach-L models. In the final stage of the proposed approach, we summarized and discussed the important findings from the convincing Mach-L models and identified the significant variables.

These algorithms and statistical analyses were executed using R programming language version 4.1.2 (R Core Team, Vienna, Austria) and implemented in RStudio version 1.1.453 [18,19] to generate all the results. These algorithms were built using the required R programming language packages, namely “randomForest,” “gbm,”

Table 2

The accuracy, sensitivity, specificity, and area under curve of the logistic regression, random forest, stochastic gradient boosting, multivariate adaptive regression spline, extreme gradient boosting, gradient boosting with categorical feature support.

	Accuracy	Sensitivity	Specificity	AUC
LR	0.5312	0.5075	0.72	0.5908
RF	0.5759	0.5327	0.92	0.7169
SGB	0.6339	0.6332	0.64	0.645
MARS	0.7902	0.8241	0.52	0.673
XGboost	0.5759	0.5578	0.72	0.6672
Catboost	0.7455	0.7739	0.52	0.6324

“earth,” “xgboost,” and “catboost.” Additionally, the “caret” R package version 6.0–90 was used to estimate the optimal hyperparameter sets for the developed effective RF, SGB, MARS, XGBoost, and CatBoost methods. The “stats” R package version 4.0.5 was used to construct the default settings for the LR model.

Results

A total of 1120 participants were included in the study. Among them, 170 participants were diagnosed with PCOS. Demographic, biochemical, and lifestyle data were grouped based on the presence or absence of PCOS (Table 1). It is worth noting that PCOS participants were younger in age but had higher levels of GOT, GPT, γ -GT, TG, and education level. Fig. 3 illustrates the age distribution of PCOS patients, with the largest number of patients (47 individuals) falling within the age range of 31.5–35 years.

Comparison results of LR and five Mach-L methods (including RF, SGB, MARS, XGBoost, and CatBoost) in predicting PCOS. It can be observed that the mean AUC of the five Mach-L methods is 0.6669, which is higher than LR's AUC of 0.5908. This indicates that the performance of these five Mach-L methods is superior to that of LR.

Table 3 illustrates the average ranking values of each risk factor generated by RF, SGB, MARS, XGBoost, and Catboost methods. From the table, it can be observed that different Mach-L methods produce varying levels of relative importance for each risk factor. To comprehensively integrate the importance rankings of all Mach-L

and LR methods for each risk factor, the average importance value for each risk factor is obtained by averaging the ranking values for each variable in each method. It is worth noting that the depth of color indicates the importance of the risk factors, where darker color represents higher importance. For example, in the RF method, the top three important factors are TG, Body fat and LDL-C.

Fig. 4 presents the risk factors described in ascending order based on their average ranking values. From the figure, it can be observed that the top 6 important risk factors for predicting PCOS are age, TG, GPT, WBC, UA, and platelet count.

Finally, Fig. 5 displays the AUC for each method. It is noticeable that LR has the lowest AUC (0.591), whereas RF achieves the highest AUC (0.717).

Discussion

In this study, by utilizing Mach-L, the six most important factors related to PCOS were identified in a group of Chinese women. These factors, listed from most important to least important, are age, TG, GPT, WBC, UA, and platelet count. It should be noted that only a limited number of similar studies could be found on PubMed. These studies either had a small sample size [20] or focused more on artificial aspects rather than clinical aspects [4]. In contrast to other research, our data also include information about participants' education, smoking, alcohol consumption, exercise, and sleep status. Although these pieces of information were not selected as important factors, this negative finding allows us to gain further insights into the role of lifestyle in PCOS.

In this study, 15.2 % of the participants were diagnosed with PCOS (170 out of 1120). This percentage is slightly higher compared to other studies. For instance, Trivax et al. reported that the prevalence of PCOS among individuals aged 15 to 44 or within the reproductive age range is between 5 % and 10 % [21]. From the age distribution, the highest prevalence of PCOS was observed in the age group of 31.5–35 years (47 cases). Compared to other studies, Ganie et al. recruited 171 suspected PCOS patients from 964 participants. The highest age group with PCOS prevalence was reported to be between 20 and 24 years [22]. In another study conducted by Alam Suha et al. it was reported that the highest

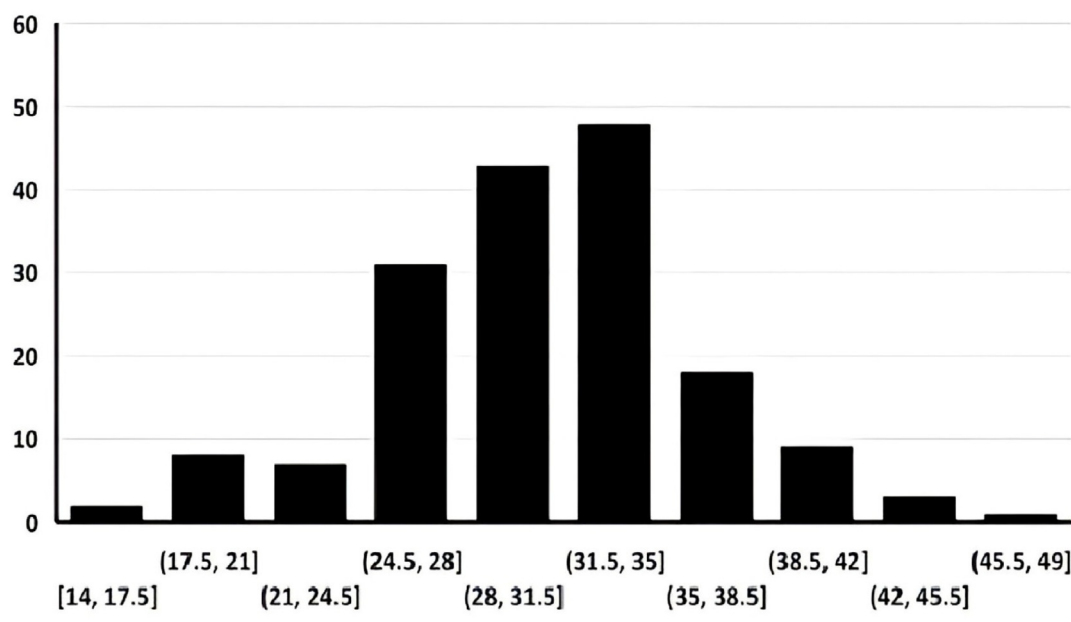


Fig. 3. Age distribution of the polycystic ovary disease patients.

Table 3
The variable importance determined by five machine learning methods.

	RF	SGB	MARS	XGboost	Catboost	MOVI
Age	78.26	51.29	100	100	100	85.91
Body fat	90.31	48.89	0	25.62	6.66	34.30
WBC	78.43	55.25	88.42	43.26	6.32	54.34
Hemoglobin	79.23	43.21	0	5.68	18.18	29.26
Platelet	100	56.2	29.55	25.64	24.56	47.19
FPG	73.66	21.07	0	12.11	32.28	27.82
GOT	70.34	16.04	42.09	37.34	48.39	42.84
GPT	75.53	100	58.81	64.12	16.25	62.94
γ-GT	76.37	47.04	0	0	8.17	26.32
Creatinine	79.54	12.58	0	0	26.08	23.64
Uric acid	88.95	61.11	47.18	18.24	23.7	47.84
Triglyceride	95.43	64.04	88.42	40.56	33.36	64.36
HDL-cholesterol	87.17	14.1	72.15	6.9	26.66	41.40
LDL-cholesterol	87.92	0	0	4.66	22.01	22.92
TSH	86.39	0	0	11.81	20.51	23.74
C-reactive protein	39.75	0	0	33.33	40.67	22.75
Education level	34.82	11.28	0	9.46	59.96	23.10
Drinking	13.9	7.47	0	0	0.22	4.32
Smoking	0	0	0	0	0	0
Sport	60.15	22.92	0	11.57	17.32	22.39
Sleep time	22.25	21.95	0	5.44	40.84	18.10

RF: random forest, SGB: stochastic gradient boosting, MARS: multivariate adaptive regression spline, XGboost: extreme gradient boosting, CatBoost: gradient boosting with categorical feature support, WBC: white blood cell count, FPG: fasting plasma glucose, GOT: serum glutamic oxaloacetic transaminase, GPT: serum glutamic pyruvic transaminase, γ-GT: γ-Glutamyl transferase, TSH: thyroid stimulating hormone, MOVI: mean of variable importance. The darker the blue color means the more important that variable is.

prevalence of PCOS was observed in the age group around 27 years [5]. Further research on larger cohorts is needed to elucidate the relationship between age and the prevalence of PCOS. Additionally, our results indicate that PCOS patients are younger in age compared to the control group (31.1 ± 5.5 years vs. 32.8 ± 7.7 years). Interestingly, there are limited studies that have focused on the age distribution of PCOS across different age groups.

The second most important factor identified by Mach-L is TG. Our data show that PCOS patients have higher TG levels compared to the control group (93.4 ± 56.8 vs. 83.7 ± 48.0 , $p < 0.05$). Our findings are consistent with other major studies. For example, the National Cholesterol Education Program (NCEP) report stated that 70 % of PCOS cases exhibit lipid abnormalities [23]. Meanwhile, Glueck et al. [24] further pointed out that 56 % of PCOS patients suffer from hypertriglyceridemia. Interestingly, PCOS patients with high TG levels also exhibit low HDL cholesterol levels. These lipid profile changes are indicative of insulin resistance [25]. The increase in TG can be explained by the following pathophysiological changes found in PCOS patients, such as obesity [26], hyperandrogenemia [27], and insulin resistance [28]. Although the complex interactions between TG and these potential disorders are intriguing, they are beyond the scope of this study.

In a retrospective study conducted by Minato et al. it was found that among 102 Japanese PCOS patients, 33.3 % had elevated liver enzyme levels [29]. Similar findings have been reported in other studies as well. The proportion of elevated liver enzymes varies from 15.4 % in Americans to 57.9 % in Italians [30,31]. In this study, it was also discovered that elevated GPT (Glutamic-Pyruvic Transaminase) levels were the third important factor for detecting PCOS. This finding is not surprising, as a meta-analysis reported that

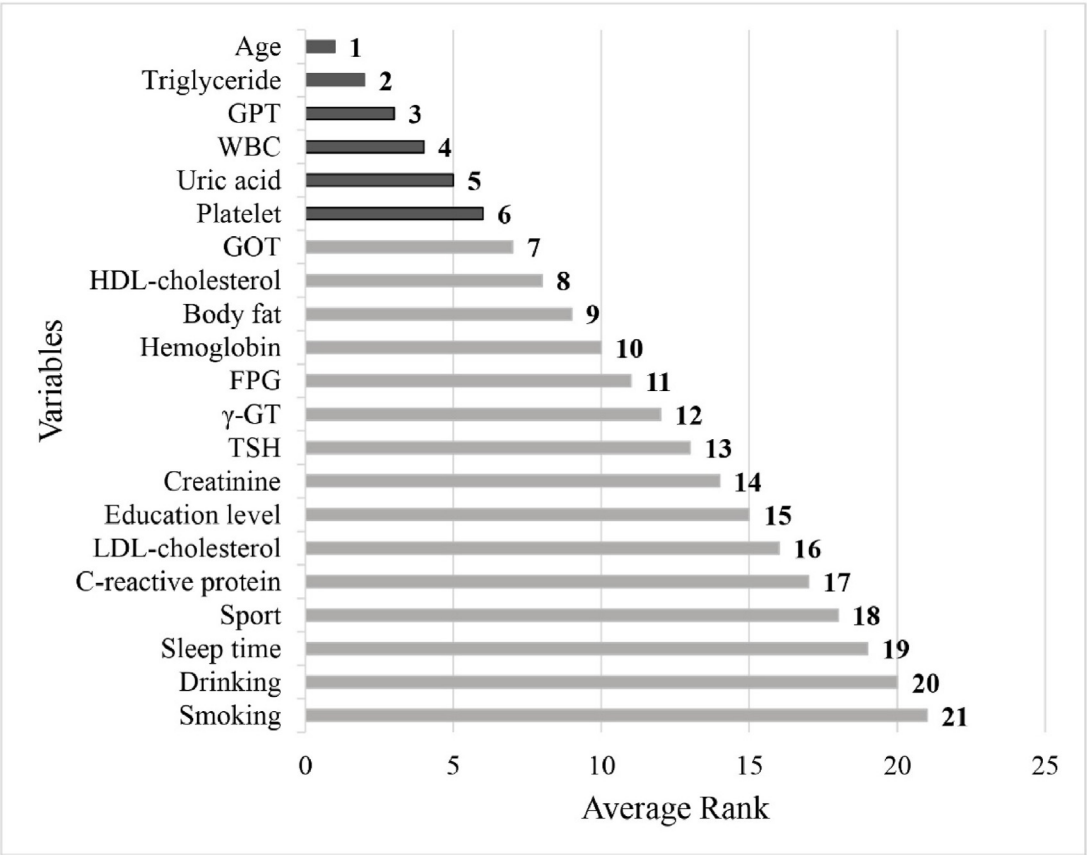


Fig. 4. Integrated importance ranking of all risk factors.

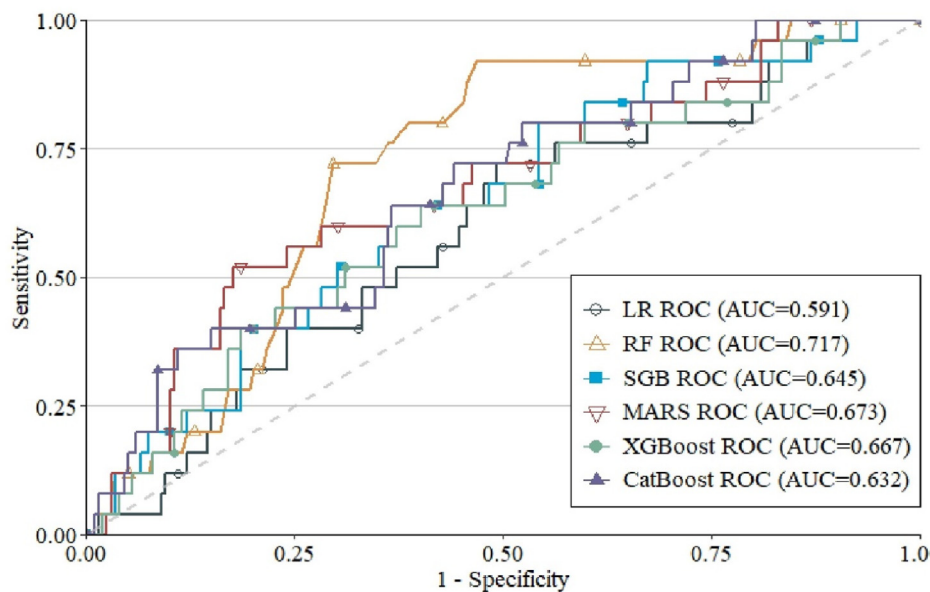


Fig. 5. The area under curve of logistic regression, random forest, stochastic gradient boosting, multivariate adaptive regression spline, extreme gradient boosting, gradient boosting with categorical feature support.

women with PCOS are four times more likely to have non-alcoholic fatty liver disease (NAFLD) [32]. The underlying pathophysiology may be attributed to the stimulation of pituitary to release FSH and LH by gonadotropin-releasing hormone, which is negatively correlated with BMI [33]. Minato et al. proposed this mechanism, which may explain the higher prevalence of NAFLD in Japanese women with PCOS [29].

Obesity, especially visceral obesity, is highly prevalent in PCOS patients [34]. It is a metabolic disorder characterized by chronic inflammation, with higher levels of pro-inflammatory cytokines, chemokines, and oxidative stress markers [35,36]. Simultaneously, PCOS is also associated with chronic inflammation. Compared to age and BMI-matched control groups, PCOS women have higher levels of C-reactive protein, interleukin-18, tumor necrosis factor, interleukin-6, and ferritin [37–39]. Therefore, it is not surprising that this study found an elevated level of WBC (White Blood Cell count) to be the fourth important factor in determining the presence of PCOS. In Orio et al. and Papalou et al. found that insulin resistance and obesity, rather than hyperandrogenism, were the main factors contributing to elevated WBC in PCOS women [39,40]. However, other authors demonstrated that androgens are also predictive factors of white blood cell count in PCOS patients [41,42]. The study found that WBC is positively correlated with total testosterone, androsterone, and DHEAS (Dehydroepiandrosterone sulfate), with testosterone being one of the main predictors of white blood cell count in multiple regression analysis [41].

Excessive secretion of androgens is one of the diagnostic criteria for PCOS. Androgens secreted by the ovaries and adrenal glands are important factors leading to the massive follicle closure, ultimately causing ovulation dysfunction in PCOS women [43]. Animal studies suggest that testosterone may increase serum UA levels by inducing hepatic metabolism of purine nucleotides [44]. Additionally, other hypothetical mechanisms have been proposed to support the relationship between them, such as oxidative stress, inflammation promotion, endothelial damage, and thrombus formation in cardiovascular diseases [45]. Therefore, in this study, UA is considered the fifth important factor for PCOS, which is entirely reasonable.

The last factor selected by Mach-L is the platelet level. Similarly, our results are consistent with other studies. For instance, in a

meta-analysis, Li et al. analyzed 866 cases of PCOS and concluded that platelet levels were significantly higher (standard mean difference = 0.43, 95 % confidence interval = 0.13–0.79) [46]. They also reported a weak positive correlation between insulin resistance measured by steady-state assessment and platelet levels. Therefore, insulin resistance is considered the cause of elevated platelet levels in PCOS patients. Further research on more patients and animals is needed to explore this relationship.

The current study has its limitations. Firstly, it is a cross-sectional study, which lacks the persuasiveness of longitudinal studies. Secondly, the AUC is not as high as in other studies. However, it is worth noting that we included information on lifestyle, such as sleep duration, exercise time, alcohol consumption, smoking, and educational level. They were not selected as important factors by Mach-L, which also provides important information that lifestyle does not play a role in the occurrence of PCOS.

Conclusion

In conclusion, compared to the traditional LR method, Mach-L demonstrated higher accuracy and specificity in predicting whether the participants in the Chinese female cohort have PCOS. The most important factor is age, followed by TG, WBC, UA, and Plt. Further investigation focusing on these factors should be emphasized.

Informed consent statement

This manuscript contains no person's details, images, or videos.

Author contributions

Conceptualization, Yao-Jen Liang and Chen-Yu Wang; Data curation, Ta-Wei Chu and Jyun-Cheng Ke; Formal analysis, Dee Pei and Chen-Yu Wang; Methodology, Jyun-Cheng Ke; Supervision, Siou-Ting Lee; Writing – original draft, Siou-Ting Lee and Jyun-Cheng Ke; Writing – review & editing, Ta-Wei Chu and Yao-Jen Liang.

Institutional Review Board statement

The study protocol was approved by the Institutional Review Board of the Tri-Service General Hospital, National Defense Medical Center (IRB No.: A202205013).

Data availability statement

Data available on request due to privacy/ethical restrictions.

Funding

The authors gratefully acknowledge financial support from the Tri-Service General Hospital, National Defense Medical Center (TSGH-D-111172).

Conflicts of interest

The authors have no conflicts of interest relevant to this article.

Acknowledgements

All authors thank all participants for the study.

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